

<Membuat Data Warehouse dan Stored Procedure untuk salah satu client ID/X Partners>

ID/X Partners – Data Engineer

Presented by

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Dedicated to creating and building reliable AI solutions from data processing to model deployment.



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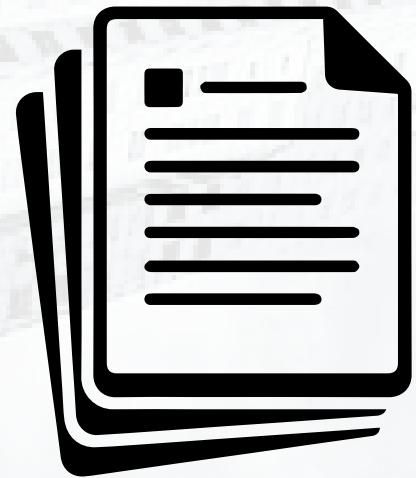
<<https://www.linkedin.com/in/fwoggo505/>>

About Company

ID/X Partners adalah perusahaan konsultan dari Indonesia yang fokus pada solusi Data Analytics, Decisioning, dan RegTech sejak 2006. Selama perjalanannya, perusahaan ini sudah membantu lebih dari 85 institusi—mulai dari bank besar, multifinance, fintech, hingga asuransi—dalam memanfaatkan analitik lanjutan dan teknologi berbasis AI. Solusi yang mereka kembangkan digunakan untuk menangani berbagai kebutuhan, seperti digital lending, pengelolaan risiko, pencegahan kejahatan finansial, hingga memastikan kepatuhan terhadap regulasi.

The logo for id/x partners, consisting of the text "id/x" in blue on a white background, followed by "partners" in white on a dark blue background.**id/x** partners

Project Portfolio



Tujuan: Membangun Data Warehouse (DWH) untuk sistem pelaporan perbankan yang lebih cepat dan terpusat.

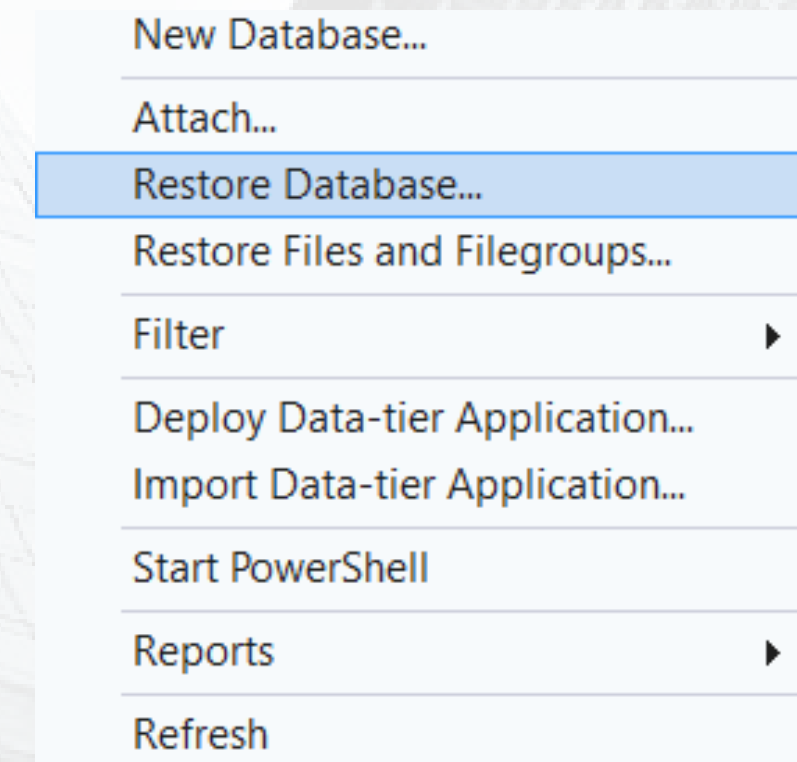


Problem: Kesulitan ekstraksi data dari berbagai format (Excel, CSV, SQL) yang menyebabkan keterlambatan laporan

Data Sources: transaction_excel, transaction_csv, dan database sample.bak (SQL Server).

Data Warehouse Creation

- Restore Database Source: Mengembalikan data histori perbankan dari file sample.bak ke dalam SQL Server.
- Database Inisialisasi: Membuat database baru bernama DWH sebagai wadah penyimpanan data pusat.
- Skema Star : Membangun 3 dimension table (DimAccount, DimCustomer, DimBranch) dan 1 fact table (FactTransaction).
- Integritas Data: Mengatur Primary Key dan Foreign Key untuk memastikan relasi antar tabel terjaga dengan baik.



Data Warehouse Creation

```
CREATE DATABASE DWH;  
GO
```

```
USE DWH;  
GO
```

```
-- DimAccount
```

```
CREATE TABLE DimAccount (  
    AccountID      INT           PRIMARY KEY,  
    CustomerID     INT           NOT NULL,  
    AccountType    VARCHAR(50)   NOT NULL,  
    Balance         BIGINT        NOT NULL,  
    DateOpened     DATE          NOT NULL,  
    Status         VARCHAR(20)   NOT NULL  
);
```

```
-- DimBranch
```

```
CREATE TABLE DimBranch (  
    BranchID       INT           PRIMARY KEY,  
    BranchName     VARCHAR(100)  NOT NULL,  
    BranchLocation VARCHAR(255)  NOT NULL  
);
```

```
-- DimCustomer
```

```
CREATE TABLE DimCustomer (  
    CustomerID     INT           PRIMARY KEY,  
    CustomerName   VARCHAR(100) NOT NULL,  
    Address        VARCHAR(255) NOT NULL,  
    CityName       VARCHAR(100) NOT NULL,  
    StateName      VARCHAR(100) NOT NULL,  
    Age            INT           NOT NULL,  
    Gender         VARCHAR(20)  NOT NULL,  
    Email          VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE FactTransaction (  
    TransactionID  INT           PRIMARY KEY,  
    AccountID      INT           NOT NULL,  
    TransactionDate DATETIME     NOT NULL,  
    Amount         BIGINT        NOT NULL,  
    TransactionType VARCHAR(50)  NOT NULL,  
    BranchID       INT           NOT NULL,
```

```
CONSTRAINT FK_Fact_Account  
    FOREIGN KEY (AccountID)  
    REFERENCES DimAccount(AccountID),
```

```
CONSTRAINT FK_Fact_Branch  
    FOREIGN KEY (BranchID)  
    REFERENCES DimBranch(BranchID)
```

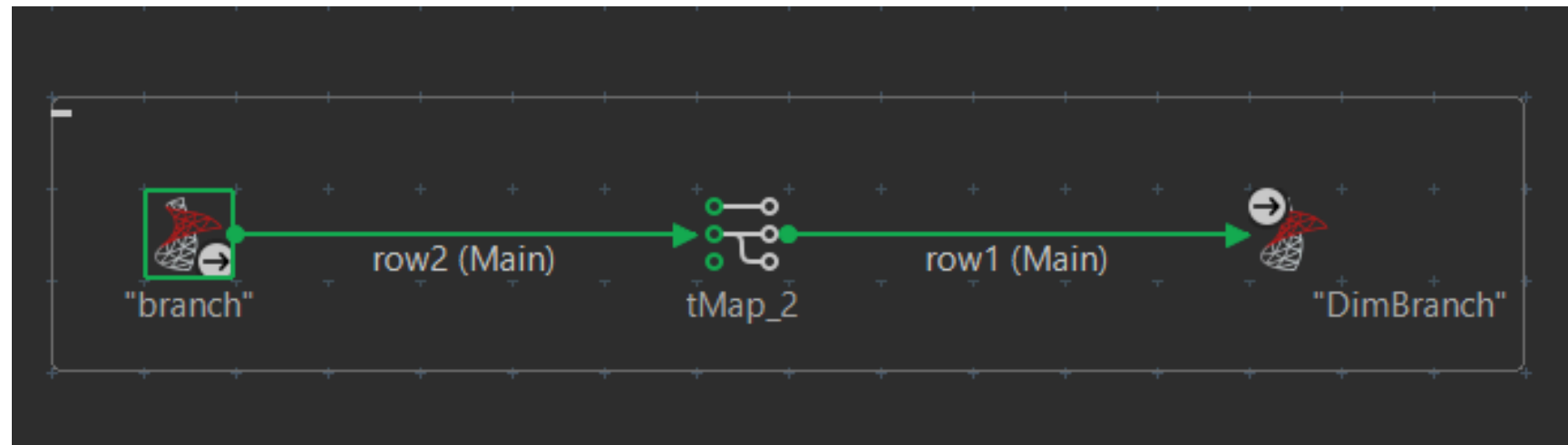
```
);
```


Create ETL Job for Dimension Table

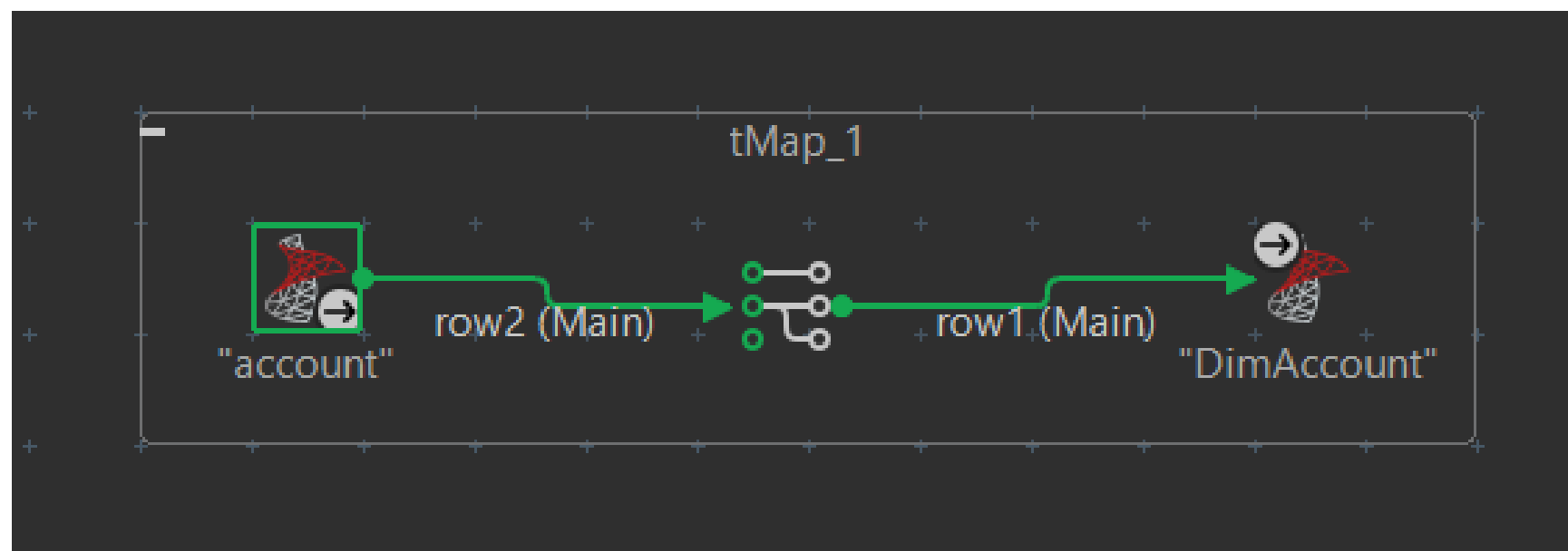
Tahap ETL (Extract, Transform, Load) menggunakan Talend untuk mengintegrasikan data dari format SQL, Excel, dan CSV ke dalam Data Warehouse secara akurat dan terstandarisasi.



Create ETL Job for Dimension Table

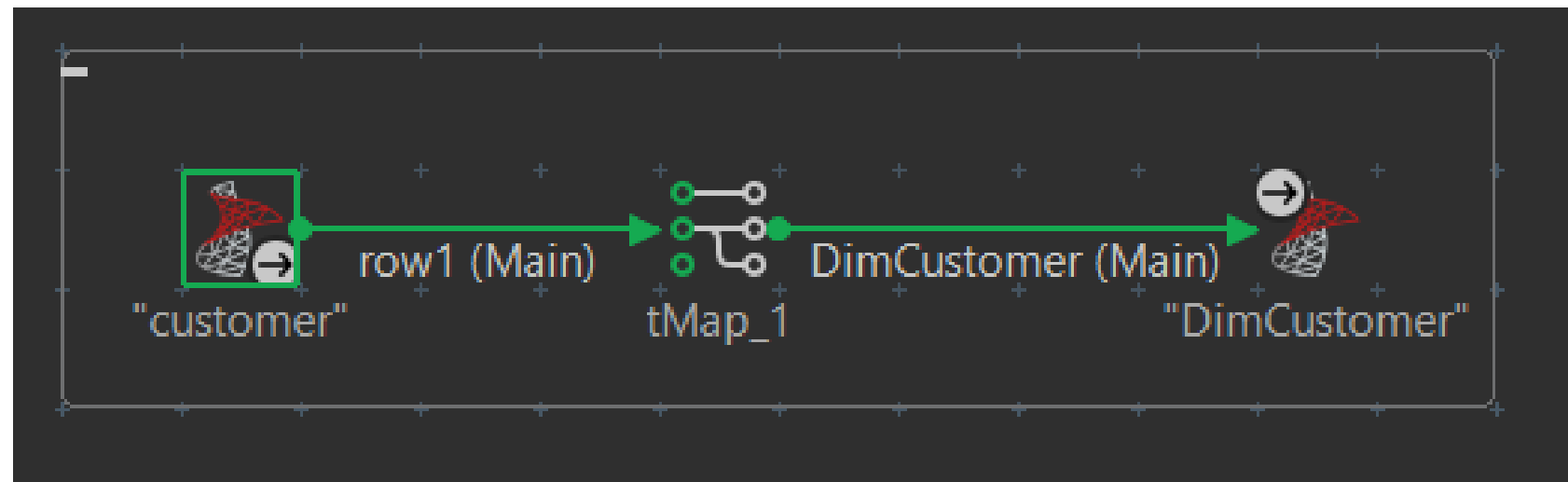


	BranchID	BranchName	BranchLocation
1	1	KC Jakarta	Jl. Gatot Subroto No 13
2	2	KC Bogor	Jl. Padjajaran No 43
3	3	KC Depok	Jl. Raya Sawangan No 34
4	4	KC Tangerang	Jl. Cisauk No 50
5	5	KC Bekasi	Jl. Ahmad Yani No 23



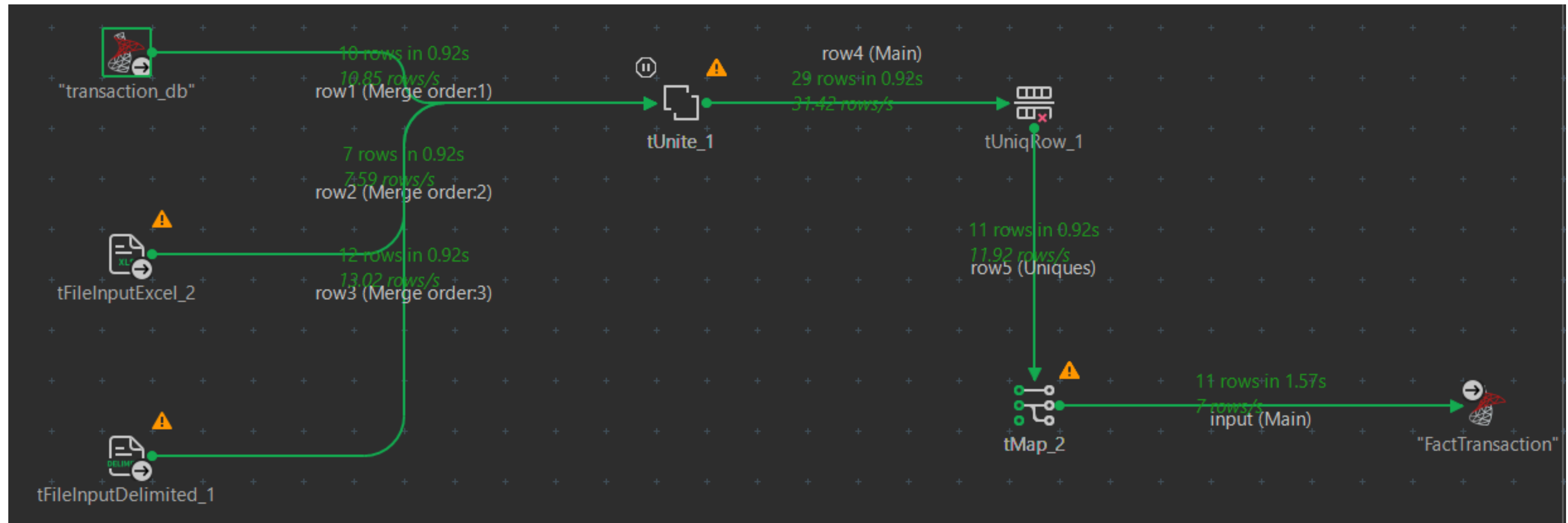
	AccountID	CustomerID	AccountType	Balance	DateOpened	Status
1	1	1	saving	1500000	2020-05-01	active
2	2	2	saving	500000	2020-06-01	active
3	3	1	checking	25000000	2020-06-21	active
4	4	3	checking	4500000	2021-06-24	terminated
5	5	4	saving	75000000	2020-06-29	active
6	6	5	checking	1500000	2020-07-01	active
7	7	6	saving	15000000	2020-07-14	terminated
8	8	7	checking	25000000	2020-07-15	active
9	9	8	saving	80000000	2020-07-15	active
10	10	9	checking	25000000	2020-07-16	active
11	11	10	saving	75000000	2020-07-24	active
12	12	11	checking	25000000	2020-08-08	active

Create ETL Job for Dimension Table



	CustomerID	CustomerName	Address	CityName	StateName	Age	Gender	Email
1	1	SHELLY JUWITA	JL. BOULEVARD NO. 31	KELAPA GADING	JAKARTA UTARA	25	FEMALE	shelly@gmail.com
2	2	BOBI RINALDO	JL. MANGGA NO. 1	TANJUNG PRIOK	JAKARTA UTARA	31	MALE	Bob@gmail.com
3	3	ADAM MALIK	JL. KINCIR ANGIN NO. 50	PADEMANGAN	JAKARTA UTARA	23	MALE	Adam@gmail.com
4	4	SUSI RAHMAWATI	JL. KENANGA NO. 11	CILANDAK	JAKARTA SELATAN	45	FEMALE	Susi@gmail.com
5	5	DIMAS PRASETYO	JL. NIAGARA NO. 69	JAGAKARSA	JAKARTA SELATAN	32	MALE	Dimas@gmail.com
6	6	AJI PANGESTU	JL. SEMPURNA NO. 2	MAMPANG PRAPATAN	JAKARTA SELATAN	40	MALE	Aji@gmail.com
7	7	BUNGA MALIKA	JL. MERAK NO. 10	CIBINONG	BOGOR	27	FEMALE	Bunga@gmail.com
8	8	RIA ADDINI	JL. ARJUNA NO. 40	MENTENG	JAKARTA PUSAT	29	FEMALE	Ria@gmail.com
9	9	LISA WULANDARI	JL. AMPERA NO. 39	TEBET	JAKARTA SELATAN	26	FEMALE	Lisa@gmail.com
10	10	RIO WIJAYA	JL. ABDUL MUIS NO. 70	GAMBIR	JAKARTA PUSAT	52	MALE	Rio@gmail.com
11	11	AHMAD RIYANSAH	Jl. KECAPU NO. 5	RARFI AN	RFKASI	27	MAI F	Ahmad@gmail.com

Create ETL Job for Fact Table



Create Stored Procedure

```
CREATE PROCEDURE DailyTransaction
    @start_date DATE,
    @end_date DATE
AS
BEGIN
    SELECT
        CAST(TransactionDate AS DATE) AS Date,
        COUNT(TransactionID) AS TotalTransactions,
        SUM(Amount) AS TotalAmount
    FROM
        FactTransaction
    WHERE
        CAST(TransactionDate AS DATE) BETWEEN @start_date AND @end_date
    GROUP BY
        CAST(TransactionDate AS DATE)
    ORDER BY
        CAST(TransactionDate AS DATE);
END;
GO

EXEC DailyTransaction @start_date = '2024-01-18', @end_date = '2024-01-20';
```

Prosedur DailyTransaction

	Date	TotalTransactions	TotalAmount
1	2024-01-18	4	11250000
2	2024-01-19	3	5400000
3	2024-01-20	1	2000000

Create Stored Procedure

```
CREATE PROCEDURE BalancePerCustomer
    @name VARCHAR(100)
AS
BEGIN
    SELECT
        c.CustomerName,
        a.AccountType,
        a.Balance,
        a.Balance + SUM(
            CASE
                WHEN f.TransactionType = 'Deposit'
                THEN f.Amount
                ELSE -f.Amount
            END
        ) AS CurrentBalance
    FROM
        DimCustomer c
    JOIN DimAccount a
        ON c.CustomerID = a.CustomerID
    JOIN FactTransaction f
        ON a.AccountID = f.AccountID
    WHERE
        a.Status = 'active'
        AND c.CustomerName LIKE '%' + UPPER(@name) + '%'
    GROUP BY
        c.CustomerName,
        a.AccountType,
        a.Balance;
END;
GO

EXEC BalancePerCustomer @name = 'Shelly';
```

Prosedur BalancePerCustomer

	CustomerName	AccountType	Balance	CurrentBalance
1	SHELLY JUWITA	checking	25000000	14000000
2	SHELLY JUWITA	saving	1500000	1600000

Link Video

https://drive.google.com/drive/folders/1WyAlywW0IjQaJ1xl-8PzWWCgwn80iD2d?usp=drive_link

Link Github Pengerjaan

<https://github.com/ratihayudianurmala/DATA-ENGINEERING>

Thank You



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